



An Evaluation of Peak Power Requirements for Full Road Transport Electrification in Relation to European Electricity System Loads

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Received: 06-15-2025

Revised: 08-20-2025

Accepted: 09-08-2025

Citation: L. Piancastelli, “An evaluation of peak power requirements for full road transport electrification in relation to European electricity system loads,” *J. Sustain. Energy*, vol. 4, no. 3, pp. 223–232, 2025. <https://doi.org/10.56578/jse040302>.



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Abstract: This study examines the implications of replacing the Italian vehicle fleet with electric vehicles powered exclusively through fast and slow charging. The purpose is to quantify the additional electrical energy and peak charging power required, and to assess their compatibility with the present characteristics of major European electricity systems. The methodology combines national mobility statistics, estimated charging demand profiles, and empirical scaling factors derived from refuelling infrastructure to determine both annual energy requirements and instantaneous power needs. The analysis indicates that full fleet electrification for night-only charging would increase national electricity consumption by approximately 40–50%, a substantial yet potentially manageable rise in annual energy consumption. By contrast, the charging power needed to support large-scale fast charging reaches values close to 280 gigawatts, far exceeding the peak loads currently managed by existing transmission networks. This peak requirement is nearly five times higher than the present Italian maximum demand and surpasses, by large margins, the peak values recorded in comparable European systems. The results indicate that the principal challenge of transport electrification lies in accommodating extremely concentrated power demand within limited temporal windows. The conclusions emphasize the need for substantial upgrades to transmission and distribution networks, complemented by the widespread adoption of controlled slow charging and demand-shifting strategies that can help reduce peak loads. These findings suggest that the feasibility of large-scale vehicle electrification hinges critically on managing instantaneous power rather than total energy, underscoring the importance of coordinating infrastructure planning, regulatory frameworks, and charging behavior to ensure that electric mobility can be integrated into existing power systems without compromising stability or reliability.

Keywords: Electric vehicle charging demand; Peak power requirements; Fast-charging infrastructure; Energy demand analysis; Grid capacity assessment; Load management strategies; Power system planning

1 Introduction

The large-scale electrification of road transport has long been recognised as a promising pathway to reduce greenhouse gas emissions, improve urban air quality, and decrease dependency on fossil fuels. However, as already discussed in the original analysis of electric-vehicle deployment and its impact on the Italian power system [1, 2], the transition from conventional fuels to electricity entails deep structural implications for both energy consumption and network operation. Recent quantitative results further reinforce these concerns, allowing for a more precise comparison with the issues previously identified. Even moderate electric-vehicle penetration—on the order of 10–20%—can significantly increase the loading of distribution transformers, modify daily load profiles, and exacerbate evening peak demand, especially under uncontrolled charging behaviour. The updated national-scale calculations confirm this tendency but reveal that a fully electric replacement of the diesel fleet would amplify these challenges to an unprecedented magnitude. The combined fast- and slow-charging demand of such a fleet amounts to approximately 104.2 TWh/yr., corresponding to a 33–40% increase relative to current Italian electricity demand or production. This aligns with the early qualitative conclusion that the energy system would need substantial reinforcement to accommodate a widespread shift toward electric mobility; however, it quantifies the scale more precisely and shows that electrifying only diesel vehicles already requires an additional annual energy volume comparable to that of several large conventional power plants operating continuously. More critically, the new analysis confirms and

extends concerns regarding power peaks, which are a fundamental bottleneck for integrating electric vehicles. While those earlier studies considered charging patterns for limited penetration scenarios, the updated computation shows that supporting nationwide fast charging for the entire diesel fleet would require an average fast-charging power of approximately 14 GW and an installed capacity of roughly 280 GW, almost five times the current Italian peak system load. This result significantly exceeds the transformer-loading margins identified in the original distribution-grid simulations, indicating that uncontrolled fast charging on a national scale would be incompatible with present-day infrastructure without substantial structural upgrades. Slow charging remains the only technically viable mechanism to limit peak stresses. The newly derived slow-charging power requirement, although significantly lower than that of fast charging, still amounts to nearly 20 GW of average nighttime load, implying a sustained increase in base demand and necessitating widespread reinforcement of residential networks. This aligns with the thesis finding that even slow, controlled charging can reshape load curves and push evening and night-time consumption into ranges not originally anticipated by existing distribution networks. The smart grid and Vehicle-to-Grid (V2G) paradigm is an enabling framework for flexible charging and bidirectional power exchange. The updated quantitative analysis reinforces the necessity of such an approach. The magnitude of the required charging infrastructure, corresponding to tens of millions of new dedicated meters, approximately matching or exceeding the number of existing domestic supply points, demonstrates that unmanaged charging would be infeasible. Instead, coordinated charging, dynamic tariffs, automated load scheduling, and V2G operation become indispensable tools to flatten demand peaks and shift aggregate consumption into periods where the system has unused capacity. This directly confirms the thesis observation that electric vehicles must evolve from passive loads to actively managed grid resources [1]. Overall, the comparison between the newly computed national-scale figures and the qualitative framework presented in the study [1] reveals strong coherence: all earlier structural concerns are not only validated but also amplified when the analysis is extended beyond partial fleet penetration to full diesel-fleet electrification. The updated numerical results clarify that the primary challenge is not annual energy production, which can be managed through gradual expansion and diversification, but rather the instantaneous power delivery required by fast charging and the infrastructural scaling needed for household charging. The integration of electric vehicles into the Italian power system, therefore, demands a paradigm in which smart-grid capabilities, controlled charging strategies, and V2G technologies evolve from optional enhancements into essential system-level requirements.

2 Structure of the Italian Circulating Vehicle Fleet and Fuel Consumption Characteristics

Table 1 provides a detailed overview of the circulating vehicle fleet in Italy, disaggregated by category and fuel type, together with indicative average fuel consumption values expressed in litres per 100 km. The table consolidates the most recent national statistics published by ANFIA (Associazione Nazionale Filiera Industria Automobilistica–National Association for the Automotive Industry Supply Chain), and ACI (Automobile Club d’Italia–Italian Automobile Club) for the year 2024, including the total number of active vehicles in each segment. These values represent the reference population used in the subsequent energy, refueling, and charging analyses presented throughout this study.

2.1 Methodology for Compiling the Vehicle Fleet and Consumption Table

The compilation of Table 1 follows a structured procedure that integrates official fleet statistics with representative fuel consumption values derived from established technical and statistical sources. The objective is to obtain coherent and comparable dataset for all vehicle categories, suitable for subsequent energy and refueling analyses. First, the total number of circulating vehicles in each segment was extracted from ANFIA and ACI national statistics for the year 2024. These data provide the official population counts for passenger cars, light trucks, medium and heavy-duty trucks, and road tractors, disaggregated by fuel type. The resulting distribution constitutes the basis for all quantitative assessments presented in this work. Second, the specific fuel consumption values reported in Table 1 were selected from recognised statistical datasets and technical studies [3–5]. For petrol and diesel vehicles, the values correspond to typical real-world performance and reflect the combined effect of engine efficiency, vehicle mass, and duty-cycle characteristics. These values are not laboratory-derived but instead represent indicative averages that can be used reliably in large-scale energy modelling. Third, for hybrid vehicles, a conservative representation was adopted. To avoid understating the liquid fuel demand of hybrid powertrains, their operation was modeled as entirely thermal. Under this assumption, the hybrid variants were assigned the same specific fuel consumption as their corresponding internal combustion counterparts. This approach ensures methodological consistency when calculating annual fuel demand, refueling frequency, and the implications of full electrification. Finally, vehicle categories with relatively small populations, such as petrol-powered road tractors or hybrid heavy-duty trucks, were retained in the table for completeness even though their contribution to national fuel consumption is negligible. Their inclusion preserves the structural symmetry of the dataset, allowing the same calculation framework to be applied uniformly across all categories. Overall, the construction of Table 2 provides a transparent and systematic foundation for estimating annual energy flows, refueling operations, and charging requirements in the context of a potential transition to a

fully electric vehicle fleet. The methodological framework adopted in this work is summarised in Figure 1, which outlines the sequence of data sources, preprocessing steps, and derived quantities used to construct the fleet and refueling table. The workflow integrates ANFIA/ACI fleet statistics, representative fuel consumption values, and category-specific tank size assumptions, and demonstrates how these inputs are systematically processed to generate the entries reported in Table 2.

Table 1. Fleet distribution, fuel consumption characteristics, and estimated annual fuel use by category

Category	Fuel	Vehicles	Cons. (L/100 km)	Fleet (L/100 km)	Fleet Share (%)	Annual Fuel (L/yr.)
Passenger cars	Petrol	17,636,755	7.8	1.3756e8	88.85	1.0303e10
Passenger cars	Hybrid petrol	1,949,658	7.8	1.5207e7	9.82	1.1390e9
Light trucks	Petrol	220,289	8.0	1.7623e6	1.14	1.3199e8
Light trucks	Hybrid petrol	29,510	8.0	2.3608e5	0.15	1.7682e7
M&H trucks	Petrol	3,358	31.1	1.0443e5	0.067	7.8218e6
M&H trucks	Hybrid petrol	12	31.1	3.73e2	0.00024	2.7952e4
Road tractors	Petrol	141	32.6	4.60e3	0.0030	3.4427e5
Road tractors	Hybrid petrol	1	32.6	3.26e1	0.000021	2.4417e3
Passenger cars	Diesel	16,896,093	9.1	1.5376e8	69.00	1.9540e10
Passenger cars	Hybrid diesel	262,273	9.1	2.3877e6	1.07	3.0344e8
Light trucks	Diesel	4,098,081	9.0	3.6883e7	16.56	4.6872e9
Light trucks	Hybrid diesel	23,827	9.0	2.1444e5	0.096	2.7252e7
M&H trucks	Diesel	719,768	31.1	2.2384e7	10.05	2.8446e9
M&H trucks	Hybrid diesel	188	31.1	5.85e3	0.0026	7.4303e5
Road tractors	Diesel	216,402	32.6	7.0565e6	3.17	8.9677e8
Road tractors	Hybrid diesel	5	32.6	1.63e2	0.000073	2.0715e4
Total petrol	-	-	-	-	100	1.1600e10
Total diesel	-	-	-	-	100	2.8300e10

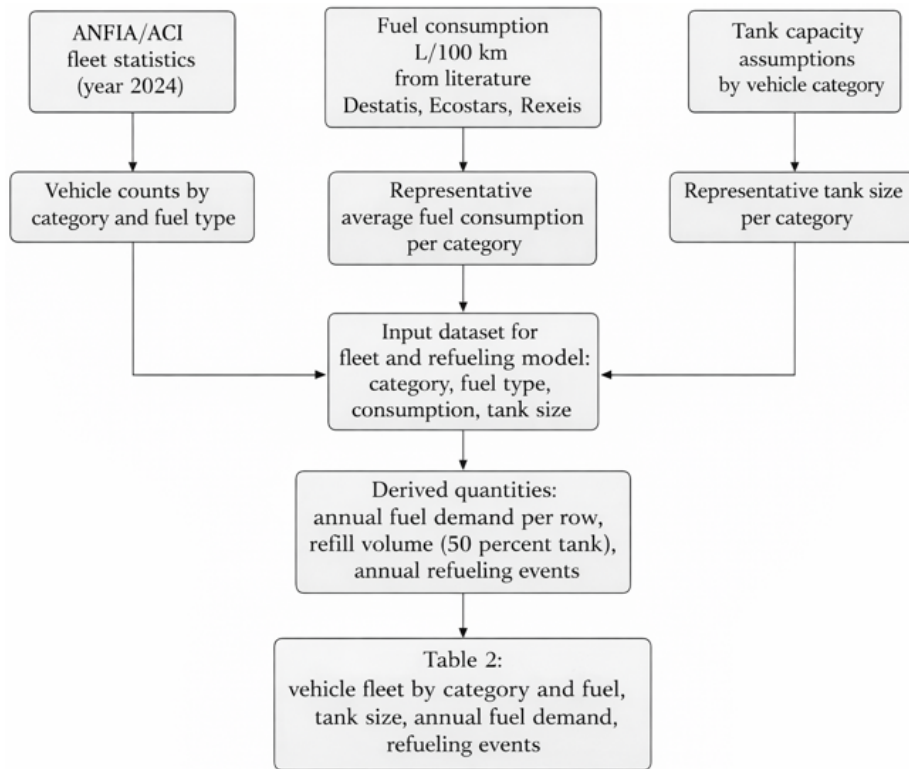


Figure 1. Workflow for constructing the fleet and refueling dataset, combining ANFIA/ACI fleet statistics (2024), literature-based fuel consumption values, and category-specific tank-size assumptions

Table 2. Vehicle fleet by category and fuel type with annual fuel demand in 2024 and computed annual refueling

Category	Fuel	Tank (L)	Annual Fuel (L)	Refill Vol. (L)	Refills/yr.	Refill Time (h/yr.)
Passenger cars	Petrol	50	1.0303e10	25	4.1212e8	3.43e7
Passenger cars	Diesel	50	1.9540e10	25	7.8160e8	6.51e7
Passenger cars	Hybrid petrol	50	1.1390e9	25	4.5560e7	3.80e6
Passenger cars	Hybrid diesel	50	3.0344e8	25	1.2138e7	1.01e6
Light trucks	Petrol	100	1.3199e8	50	2.6398e6	2.20e5
Light trucks	Diesel	100	4.6872e9	50	9.3744e7	7.81e6
Light trucks	Hybrid petrol	100	1.7682e7	50	3.5364e5	2.95e4
Light trucks	Hybrid diesel	100	2.7252e7	50	5.4504e5	4.54e4
M&H trucks	Petrol	300	7.8218e6	150	5.2145e4	4.35e3
M&H trucks	Diesel	300	2.8446e9	150	1.8964e7	1.58e6
M&H trucks	Hybrid petrol	300	2.7952e4	150	1.8635e2	1.55e1
M&H trucks	Hybrid diesel	300	7.4303e5	150	4.9535e3	4.13e2
Road tractors	Petrol	750	3.4427e5	375	9.1806e2	7.65e1
Road tractors	Diesel	750	8.9677e8	375	2.3913e6	1.99e5
Road tractors	Hybrid petrol	750	2.4417e3	375	6.5112	5.43e1
Road tractors	Hybrid diesel	750	2.0715e4	375	5.5239e1	4.60e0
Total	-	-	3.99e10	-	1.334e9	1.11e8

2.2 Method for Computing Annual Refueling Events by Vehicle Category

The annual number of refueling events for each vehicle category is determined by combining the allocated fuel consumption with the representative tank capacity. Based on national fuel statistics for Italy in 2024, the total gasoline consumption amounts to approximately 11.6 billion liters out of a combined gasoline and diesel consumption of about 39.9 billion liters. The total fuel consumption is distributed among vehicle categories proportionally to the number of vehicles and their average fuel consumption. For each category i , the annual fuel consumption F_i (L/yr.) is thus obtained.

Assuming that each refueling restores, on average, 50% of the tank capacity, the effective fuel volume delivered per refueling is:

$$V_i = 0.5T_i \quad (1)$$

where, T_i is the tank capacity (L). The annual number of refueling events is:

$$N_i^{ref} = \frac{F_i}{0.5T_i} \quad (2)$$

where, N_i^{ref} is the annual number of refueling events for category i . This formulation explicitly states that refueling frequency depends on both annual fuel consumption and tank size. Categories with large fuel demand and small tanks exhibit a high refueling frequency, whereas heavy-duty categories with large tanks show reduced event frequency despite substantial annual fuel use. After computing N_i^{ref} for each class, the total annual number of refueling events is obtained by summation:

$$N_{tot} = \sum_i N_i^{ref} \quad (3)$$

Finally, the total annual refueling time is estimated by multiplying the number of refueling events by the assumed average refueling duration of 5 minutes:

$$T_{ref} = N_{tot} \frac{5}{60} h \quad (4)$$

where, 5 minutes is the assumed average duration of a refueling event, and T_{ref} is the total annual refueling time expressed in hours. This procedure is applied consistently to both diesel and gasoline fleets, using the category-specific tank capacities reported in Table 2.

3 Annual Diesel and Gasoline Refueling Events and Associated Time Requirements

Using the methodology described in Section 2.2, the total number of annual refueling events for the Italian vehicle fleet N_{tot} is estimated as:

$$N_{tot} = 1.334 \times 10^9 \quad (5)$$

This corresponds to a total annual refueling time:

$$T_{ICE} \approx 1.11 \times 10^8 \text{ h/yr.} \quad (6)$$

These values provide the baseline for comparison with the electrified fleet scenario.

4 Estimated Annual Charging Time for a Fully Electric Vehicle Fleet

To evaluate the impact of full fleet electrification, the total number of refueling events is used as a baseline for estimating charging requirements. Assuming that battery electric vehicles (BEVs) have an effective driving range equal to 50% of that of conventional vehicles, the number of required energy stops N_{EV} doubles:

$$N_{EV} = 2N_{tot} = 2.668 \times 10^9 \quad (7)$$

Assuming that 50% of charging events occur in fast-charging mode, the number of fast-charging sessions is:

$$N_{fast} = 0.5N_{EV} = 1.334 \times 10^9 \quad (8)$$

Assuming a representative fast-charging duration of 1 hour per event, the total annual charging time T_{EV} becomes:

$$T_{EV} = N_{fast} \times 1\text{h} = 1.334 \times 10^9 \text{ h/yr.} \quad (9)$$

This value exceeds the current refueling time by more than one order of magnitude.

4.1 Increase in Time Lost and Associated Societal Impact for Battery Electric Vehicle Users

The additional time required under full electrification is:

$$\Delta T = T_{EV} - T_{ICE} = 1.334 \times 10^9 - 1.11 \times 10^8 \approx 1.22 \times 10^9 \text{ h/yr.} \quad (10)$$

To provide a clearer interpretation, the time difference may be expressed in equivalent human working lifetimes. Using a notional human working-life duration:

$$L_{life} = 40\text{yr.} \approx 7.0 \times 10^4 \text{ h} \quad (11)$$

The aggregate annual time lost corresponds to 17,500 human lifetimes per year:

$$\frac{\Delta T}{L_{life}} = \frac{1.22 \times 10^9}{7.0 \times 10^4} \approx 1.75 \times 10^4 \quad (12)$$

Even when interpreted in terms of continuous (non-working) time, the additional time burden amounts to:

$$\Delta T_{yr} = \frac{1.22 \times 10^9}{24 \times 365} \approx 1.40 \times 10^5 \text{ yr} \quad (13)$$

It equivalents to more than 140,000 years of uninterrupted human time. This magnitude illustrates the substantial societal cost associated with the temporal dimension of fast charging, particularly when aggregated over the entire fleet. From an economic perspective, the lost time represents a direct reduction in productive capacity, increased opportunity cost for individuals, and a non-trivial burden on logistics, transport efficiency, and overall system performance. These temporal externalities should therefore be explicitly accounted for in large-scale fleet electrification planning, as they constitute an operational constraint largely absent in the fossil-fuel refueling paradigm.

4.2 Fast-Charging Energy Demand

Fast charging is assumed to restore 50% of the battery capacity per event, with an efficiency of 0.88. The electrical energy E_i^{grid} drawn from the grid during a single fast-charging event for category i is:

$$E_i^{grid} = \frac{C_i}{0.88} \quad (14)$$

where, C_i is the representative battery capacity in kWh. The annual fast-charging energy E_i^{fast} demand is:

$$E_i^{fast} = \frac{N_i^{fast} E_i^{grid}}{1000} \text{MWh} \quad (15)$$

where, N_i^{fast} is the annual number of fast-charging events. Table 3 and Figure 2 summarize the resulting values for the BEV fleet.

The total fast-charging E_{tot}^{fast} energy demand approximately amounts to:

$$E_{tot}^{fast} = 5.41 \times 10^7 \text{MWh} = 54.1 \text{TWh/yr.} \quad (16)$$

Table 3. Annual fast-charging energy demand by vehicle category and fuel type for the full fleet, assuming 50% of charging events occur in fast-charge mode and charging efficiency of 0.88

Category	Fuel	Battery (kWh)	Grid Energy per Fast Charge (kWh)	Annual Fast-Charge Demand (MWh)
Passenger cars	Petrol	60	68.18	1.41e7
Passenger cars	Diesel	60	68.18	2.67e7
Passenger cars	Hybrid petrol	60	68.18	1.55e6
Passenger cars	Hybrid diesel	60	68.18	4.14e5
Light trucks	Petrol	90	102.27	1.35e5
Light trucks	Diesel	90	102.27	4.79e6
Light trucks	Hybrid petrol	90	102.27	1.81e4
Light trucks	Hybrid diesel	90	102.27	2.79e4
M&H trucks	Petrol	450	511.36	1.33e4
M&H trucks	Diesel	450	511.36	4.85e6
M&H trucks	Hybrid petrol	450	511.36	4.76e1
M&H trucks	Hybrid diesel	450	511.36	1.27e3
Road tractors	Petrol	600	681.82	3.13e3
Road tractors	Diesel	600	681.82	8.16e5
Road tractors	Hybrid petrol	600	681.82	2.22e1
Road tractors	Hybrid diesel	600	681.82	1.89e2
Total	-	-	-	5.41e7

4.3 Slow-Charging Energy Demand

The remaining 50% of charging events are assumed to take place overnight, with no time cost and a representative charging efficiency of 0.95. Since both fast and slow charging supply the same useful battery energy, the total useful energy is:

$$E_{tot}^{slow} = \frac{E_{useful}^{fast}}{0.95} \quad (17)$$

where, the useful fast-charging energy is:

$$E_{useful}^{fast} = 0.88 E_{tot}^{fast} = 0.88 \times 54.1 \text{TWh/yr.} \approx 47.6 \text{TWh/yr.} \quad (18)$$

Thus, the total slow-charging energy demand is:

$$E_{tot}^{slow} = \frac{47.6}{0.95} \approx 50.1 \text{TWh/yr.} \quad (19)$$

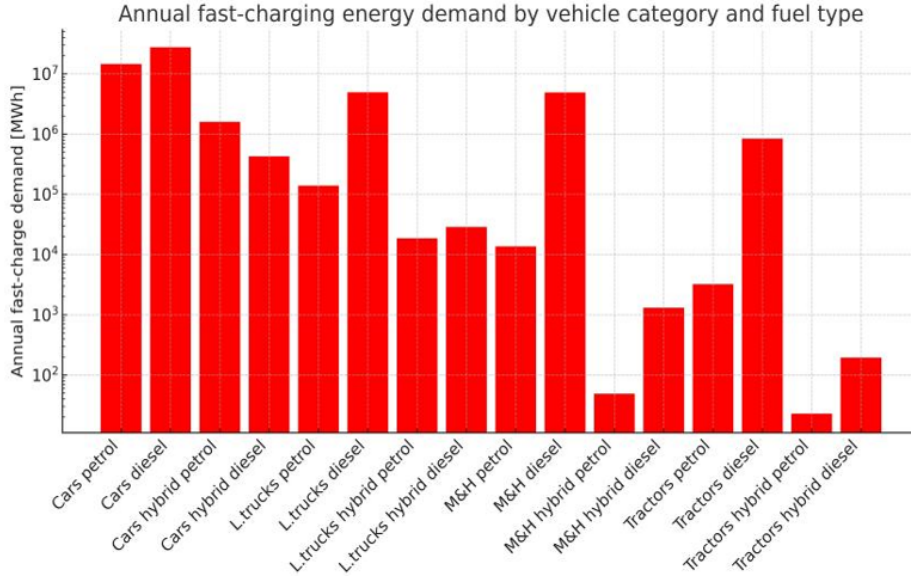


Figure 2. Simulated annual fast-charging demand by vehicle category and fuel type in 2024

4.4 Total National Battery Electric Vehicle Energy Demand and Increase in Annual Electrical Energy Demand

The total energy demand associated with a fully electric replacement of the diesel vehicle fleet is the sum of the fast-charging and slow-charging components:

$$E_{BEV,tot} = E_{tot}^{fast} + E_{tot}^{slow} = 104.2 \text{ TWh/yr.} \quad (20)$$

When compared with national electricity production in 2024 (264 TWh), this additional energy requirement represents:

$$\frac{104.2}{264} \approx 0.40 \quad (21)$$

That is, an increase of approximately 40% in current Italian electricity production. Relative to national electricity demand (312 TWh), the required increment is:

$$\frac{104.2}{312} \approx 0.33 \quad (22)$$

It correspondings to roughly 33% of present electricity consumption. These results indicate that full electrification of the diesel fleet would require an increase of approximately one-third to 40% in the annual volume of electricity that must be generated, imported, or otherwise managed by the national power system.

4.5 Average and Peak Charging Power Requirements

Assuming that fast-charging infrastructure operates for 10.5 hours per day, the annual operating time:

$$T_{open} = 10.5 \times 365 \approx 3833 \text{ h/yr.} \quad (23)$$

The corresponding average charging power is:

$$P_{avg} = \frac{E_{tot}^{fast}}{T_{open}} \approx 1.4 \times 10^4 \text{ MW} \quad (24)$$

Using the empirical observation that installed refueling power is roughly twenty times the average delivered power, the installed fast-charging capacity required nationally is:

$$P_{inst} \approx 20 P_{avg} \approx 2.80 \times 10^5 \text{ MW} = 280 \text{ GW} \quad (25)$$

5 Comparison Between Required Battery Electric Vehicle Charging Capacity and the Current Italian Electricity System

This section compares the annual electrical energy demand and the charging power requirements of a fully electrified diesel vehicle fleet with the present capabilities of the Italian electricity system. The analysis is based on the mean annual 2024 national statistics reported by Terna [6], which indicate a net electricity production of approximately 264 TWh and a total national demand of about 312 TWh. The peak system load in 2024 reached values of nearly 60 GW.

5.1 Implications for the National Grid

The comparison between the required BEV charging capacity and the present electricity system indicates two distinct stress mechanisms. First, the annual energy requirement of 104.2 TWh represents a roughly 33–40% increase in national electricity production or procurement. Second, and more critically, the charging power capacity required to sustain nationwide fast charging exceeds the current peak system load by nearly five. This large mismatch reflects the fundamental difference between centralized, continuous electricity generation and the highly clustered, time-concentrated nature of vehicle fast charging. These findings suggest that large-scale fleet electrification would require not only substantial expansion of generation capacity but also major reinforcement of transmission and distribution infrastructure, especially in areas with dense vehicle traffic or heavy reliance on long-haul transport. Furthermore, the disproportionate impact of fast charging on peak demand highlights the importance of demand-shifting strategies, such as controlled slow charging, workplace charging, and dynamic tariffs, in reducing peak loads and enhancing the feasibility of operating a fully electric national vehicle fleet.

5.2 Comparison with National Peak Electrical Loads

Figure 3 compares the peak charging power required to sustain nationwide fast charging of a fully electric diesel vehicle fleet (approximately 280 GW) with the maximum electrical loads recorded in 2024 by the Italian, Spanish, German, and French power systems [7–13]. The comparison shows that the fast-charging peak power demand would exceed the Spanish system peak by more than one order of magnitude and would represent nearly five times the Italian peak load, four times the German peak load, and more than three times the French winter peak.

This quantitative imbalance highlights that the primary constraint associated with large-scale transport electrification is not the annual energy volume to be produced, but rather the instantaneous power that the system must be capable of delivering during hours of intense charging activity. Addressing this challenge would require substantial reinforcement of transmission and distribution infrastructure, a widespread deployment of controlled slow-charging solutions, and advanced demand-side management strategies, including demand response mechanisms [14–20].

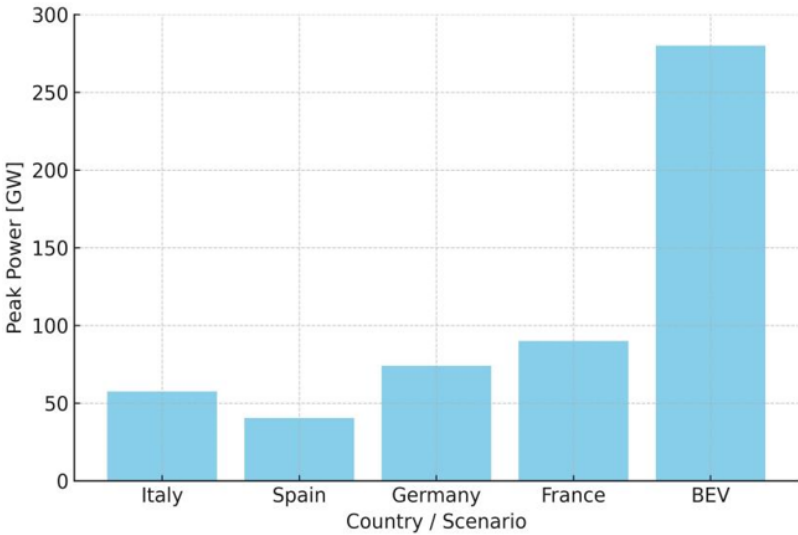


Figure 3. Comparison between battery electric vehicle (BEV) fast-charging required peak power and observed national peak electrical loads in 2024

5.3 Implications of Large-Scale Slow Charging

The feasibility of electrifying the entire Italian vehicle fleet depends not only on the total annual electrical energy required but also on the temporal distribution of charging demand. Fast charging imposes very high instantaneous

power requirements, whereas slow charging yields a considerably more manageable load profile for the power system. Assuming a daily charging event, an eight-hour night-time window, and a charging efficiency of 0.95, slow charging reduces the annual energy withdrawn from the grid by roughly 8% compared to fast charging. Distributing this energy across the assumed nighttime window results in an average additional load of approximately 20 GW. Although markedly lower than the instantaneous requirements associated with nationwide fast charging, this still represents a substantial share of current Italian peak demand and would materially reshape the national load curve. Widespread slow charging also implies significant infrastructure expansion. Each vehicle would require a dedicated metering and supply point, yet Italy currently has around 41 million passenger cars and only about 36 million existing electrical delivery points. Scaling slow charging to the national fleet would therefore require an increase of at least 100% in the number of supply points, including those for commercial vehicles. Such growth would constitute a major enlargement of the low-voltage distribution network. Overall, while slow charging alleviates the extreme peak loads associated with fast charging, it still produces substantial and geographically concentrated increases in base demand. Large-scale electrification will therefore require coordinated grid reinforcement and the adoption of controlled charging strategies to ensure that slow charging can be integrated without compromising system stability or reliability.

6 Conclusions

This study assessed the additional electrical energy and charging power required to electrify the entire Italian vehicle fleet and compared these requirements with the current operating characteristics of major European electricity systems. The analysis indicates that full fleet electrification would result in a substantial increase in national electricity consumption, yet one that remains broadly compatible with the long-term expansion of generation capacity. In contrast, the instantaneous power needed to support widespread fast charging reaches values far above the peak loads currently sustained by existing transmission and distribution networks. The comparison with observed system peaks in Italy, Spain, Germany, and France indicates that the charging power associated with a fully electric road transport sector exceeds present national load maxima by wide margins. This gap highlights a structural limitation inherent in the temporal concentration of charging demand, which imposes stress on the power system that cannot be mitigated solely through increases in energy production. The results demonstrate that the main challenge of large-scale vehicle electrification is not the total annual energy volume, but rather the grid's ability to accommodate high, spatially and temporally clustered power demand. Addressing this issue will require coordinated investments in network reinforcement, the deployment of controlled slow-charging solutions, and the adoption of advanced demand management strategies that can shift or smooth charging profiles. These measures are essential to ensure that the electrification of the Italian vehicle fleet can be integrated into the national power system without compromising operational security or system reliability.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

References

- [1] M. Muratori, "Impact of uncoordinated plug-in electric vehicle charging on residential power demand," *Nat. Energy*, vol. 3, no. 3, pp. 193–201, 2018. <https://doi.org/10.1038/s41560-017-0074-z>
- [2] E. Pierri, V. Cirillo, T. Vietor, and M. Sorrentino, "Adopting a conversion design approach to maximize the energy density of battery packs in electric vehicles," *Energies*, vol. 14, no. 7, p. 1939, 2021. <https://doi.org/10.3390/en14071939>
- [3] Statistisches Bundesamt (Destatis), "Mileage and fuel consumption of households' cars," 2025. <https://www.destatis.de/EN/Themes/Society-Environment/Environment/Environmental-Economic-Accounting/transport-tourism/Tables/mileage-fuel-consumption.html>
- [4] European Environment Agency (EEA), "CO₂ emissions performance of new passenger cars in Europe," 2024. <https://www.eea.europa.eu/en/analysis/indicators/co2-performance-of-new-passenger>
- [5] M. Rexeis, M. Röck, and S. Hausberger, "Comparison of fuel consumption and emissions for representative heavy-duty vehicles in Europe," Fact Sheet, International Council on Clean Transportation (ICCT), 2018. <https://theicct.org/publications/HDV-EU-fuel-consumption-and-emissions-comparison>
- [6] Terna - Rete Elettrica Nazionale Società per Azioni, "Consumi elettrici in Italia nel 2024," press release and statistical data, Terna S.p.A., 2025.

- [7] Terna - Rete Elettrica Nazionale Società per Azioni, “Electricity consumption increases by 2.2% in 2024,” press release, Terna S.p.A., Rome, Italy, 2025. <https://www.terna.it/en/media/press-releases/detail/electricity-consumption-2024>
- [8] Service des données et études statistiques (SDES), “Bilan énergétique de la France en 2024 – données provisoires,” Ministère de la Transition écologique et de la Cohésion des territoires, Paris, France, 2025. <https://www.statistiques.developpement-durable.gouv.fr/bilan-energetique-de-la-france-en-2024-donnees-provisaires>
- [9] Arbeitsgemeinschaft Energiebilanzen (AGEB), “Energieverbrauch in Deutschland im Jahr 2024,” preliminary data, Berlin, Germany, 2025. <https://ag-energiebilanzen.de/energieverbrauch-in-deutschland/>
- [10] Red Eléctrica de España (REE), “Informe del Sistema Eléctrico Español 2024,” Madrid, Spain, 2025. <https://www.ree.es/es/datos/publicaciones/informe-del-sistema-electrico-espanol>
- [11] E. Lorenzini, O. Manca, R. Mastrullo, and S. Nardini, “Selected papers from the AIGE 2016 conference on energy conversion, management, recovery, saving, storage and renewable systems,” *Heat Transfer Eng.*, vol. 41, no. 12, pp. 1011–1013, 2020. <https://doi.org/10.1080/01457632.2019.1600861>
- [12] L. Piancastelli, “CFD-based analysis and optimization of depressurization systems for engine test benches of small propeller aircraft,” *Power Eng. Eng. Thermophys.*, vol. 4, no. 2, pp. 110–120, 2025. <https://doi.org/10.56578/peet040203>
- [13] L. Piancastelli, “Human-factors and power-system implications of consumer electronics-oriented vehicle design and full-electric fleet deployment,” *Power Eng. Eng. Thermophys.*, vol. 4, no. 3, pp. 168–177, 2025. <https://doi.org/10.56578/peet040303>
- [14] Ministero dell’Ambiente e della Sicurezza Energetica (MASE), “Bollettino petrolifero,” Statistiche energetiche e minerarie (SISEN), Roma, Italy, 2024. <https://sisen.mase.gov.it/dgsaie/bollettino-petrolifero>
- [15] P. Richardson, D. Flynn, and A. Keane, “Impact assessment of varying penetrations of electric vehicles on low voltage distribution systems,” in *IEEE Power and Energy Society General Meeting*, Minneapolis, MN, USA, 2010, pp. 1–6. <https://doi.org/10.1109/PES.2010.5589940>
- [16] K. Clement-Nyns, E. Haesen, and J. Driesen, “The impact of charging plug-in hybrid electric vehicles on a residential distribution grid,” *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 371–380, 2010. <https://doi.org/10.1109/TPWRS.2009.2036481>
- [17] S. Deilami, A. S. Masoum, P. S. Moses, and M. A. S. Masoum, “Real-time coordination of plug-in electric vehicle charging in smart grids to minimize power losses and improve voltage profile,” *IEEE Trans. Smart Grid*, vol. 2, no. 3, pp. 456–467, 2011. <https://doi.org/10.1109/TSG.2011.2159816>
- [18] J. Hu, S. You, M. Lind, and J. Østergaard, “Coordinated charging of electric vehicles for congestion prevention in the distribution grid,” *IEEE Trans. Smart Grid*, vol. 5, no. 2, pp. 703–711, 2014. <https://doi.org/10.1109/TSG.2013.2279007>
- [19] M. Yilmaz and P. T. Krein, “Review of battery charger topologies, charging power levels, and infrastructure for plug-in electric and hybrid vehicles,” *IEEE Trans. Power Electron.*, vol. 28, no. 5, pp. 2151–2169, 2013. <https://doi.org/10.1109/TPEL.2012.2212917>
- [20] T. R. Hawkins, B. Singh, G. Majeau-Bettez, and A. H. Strømman, “Comparative environmental life cycle assessment of conventional and electric vehicles,” *J. Ind. Ecol.*, vol. 17, no. 1, pp. 53–64, 2013. <https://doi.org/10.1111/j.1530-9290.2012.00532.x>